

Page-Curl Shader

1. Progress-to-Angle Mapping

A single scalar $p \in [0, 1]$ drives the turn. The spine angle φ is

$$\varphi(p) = \begin{cases} -\pi p & \text{forward turn,} \\ -\pi(1-p) & \text{reverse turn.} \end{cases}$$

2. Page Curl (vertex shader)

Each vertex along the page width is parameterised by $t = x/w$, where $t \in [0, 1]$ (0 at spine, 1 at free edge). The per-vertex rotation angle is

$$\phi(t) = \varphi + A \cdot t \cdot \sin(2\varphi), \quad A = 0.4 \text{ (bend amount).}$$

The vertex position is then

$$x' = x \cos \phi, \quad z' = -x \sin \phi.$$

The $\sin(2\varphi)$ envelope naturally provides the correct curl direction: positive during the lift phase ($\varphi \in [0, -\pi/2]$), zero at vertical, and negative during the fall phase ($\varphi \in [-\pi/2, -\pi]$).

Velocity constraint — the edge must never reverse direction:

$$\frac{d\phi_{\text{edge}}}{d\varphi} = 1 + 2A \cos(2\varphi) \geq 1 - 2A > 0 \implies A < 0.5.$$

With $A = 0.4$: minimum edge velocity = $0.2 > 0$.

3. Physics Settle (after drag release)

Progress p evolves under constant-acceleration gravity plus linear drag:

$$\dot{v} = \text{dir} \cdot G - D \cdot v, \quad p_{n+1} = \text{clamp}(p_n + v \Delta t, 0, 1),$$

where $G = 5.0$, $D = 6.5$, $\text{dir} = \text{sign}(\text{target} - p)$, $\Delta t = 1/60$.

Velocity is seeded at release with the drag momentum, clamped to a minimum of $|v_0| \geq 0.8$ to prevent stalling at the midpoint (where the bend envelope is also zero).

4. Timed Animation (buttons / keyboard)

Linear progress:

$$p(t) = \frac{t}{T}, \quad T = 1.2 \text{ s.}$$

5. Crease Shadow

The crease-line opacity follows a bell curve peaking at mid-turn:

$$\alpha_{\text{crease}} = \sin(\pi p).$$